# Document to Proteomics Analysis R

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## Parameters overview:

Parameters should be imported from metadata or pre-defined configuration file for data processing:

| Step | Name | Type | Description | Function |
|---|---|---|---|---|
| Sample Filter | nSample | int | The count of samples with valid data should present in a condition | MissingValueFilter() |
| | nCondition | int | The count of conditions should have data | |
| | nUniPep | int | The count of unique peptide should present in a protein | |
| Normalization | method | char | The name of normalization method, currently only supports “Median” | ProteinNormalization() |
| | species | char | The reference species used for normalization | |
| Imputation | method | chr | The name of imputation method, currently supports 3 methods: "Min", "NormDist_Fixed", "NormDist_Quantile". | Log2AndImputation() |
| | par | dbl | The corresponding parameter to the imputation method. | |
| Statistic Test | workflow_text, | char | A customized label for processing workflow, eg. “Method1” | TwoSampleTest() |
| | equal_var | lgl (boolean) | Assume equal variance of samples. If TRUE, will use student t-test; If FALSE, will use welchs t-test. | |

## Functions

- AddMissingValue(df)

Add the missing samples. The default SN “BSG factory Report” is a long table containing only the proteins/peptides with quantity, and the missing samples are absent.

**Parameters:**

| Name | Type | Description | Default |
|---|---|---|---|
| df | data.table or data.frame | The standardised protein table | required |

- MultiGeneAnnotation(ProteinAccessions, fasta_map)

To annotate multi ProteinAssessions containing ";"

**Parameters:**

| Name | Type | Description | Default |
|---|---|---|---|
| ProteinAccessions | vector of character | The column of protein accession to be annotated | required |
| fasta_map | data.table or data.frame | The imported fasta map | required |

- GetProteinMatrix(df, fasta_map)

Process raw data (default BSG factory report) into standardised protein table

**Parameters:**

| Name | Type | Description | Default |
|---|---|---|---|
| df | data.table or data.frame | The imported BSG factory report | required |
| fasta_map | data.table or data.frame | The imported fasta map | required |

- ProteinNormalization(df, method, species)

Normalize the protein quantity based on given method and species. Currently only supports “Median” normalization. Species should be upper-case, like “HUMAN”, “YEAST”.

**Parameters:**

| Name | Type | Description | Default |
|---|---|---|---|
| df | data.table or data.frame | The standardised protein table | required |
| method | char | The name of normalization method, currently only supports “Median” | required |
| species | char | The reference species used for normalization | required |

■ Log2AndImputation(df, method, par)

Log2 transform the protein quantity and impute the missing value by give method: "Min", "NormDist_Fixed", "NormDist_Quantile".

**Parameters:**

| **Name** | **Type** | **Description** | **Default** |
|---|---|---|---|
| df | data.table or data.frame | The standardised protein table | required |
| method | char | The name of imputation method | required |
| par | dbl | The corresponding parameter to the imputation method. | required |

■ TwoSampleTest(df, workflow_text, equal_var = TRUE)

Perform two sample comparison and return t test and Limma linear-model test results.

**Parameters:**

| **Name** | **Type** | **Description** | **Default** |
|---|---|---|---|
| df | data.table or data.frame | The standardised protein table | required |
| workflow_text | char | A customized label for processing workflow, eg. “Method1” | required |
| equal_var | lgl (Boolean) | Assume equal variance of samples. If TRUE, will use student t-test; If FALSE, will use welchs t-test. | TRUE |